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Bruno Gomes Coelho

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in/bruno-coelho-277519129/

Education

- **New York University (NYU)** — NY, USA
  PhD in Computer Science focused on Machine Learning — *September 2020 – May 2025*

- **New York University (NYU)** — NY, USA
  Master's in Computer Science — *September. 2020 – May 2022*

- **University of São Paulo (USP), ICMC** — SP, Brasil
  Bachelor's Degree in Computer Science; Emphasis in Data Science; GPA: 8.8/10.0 — *February 2016 – January 2020*

Industry Experience

- **Kensho - A company of S&P Global** — NY, USA
  Applied Scientist — *June 2025 - current*

  - **Focus**: Text-to-SQL models, RL, evaluation and benchmarking agentic systems.

  Machine Learning Intern (NLP Team) — *Summer 2023 & 2024 ; February-May 2025*

  - **2025**: Research on new methodologies for creating and evaluating RAG systems with unstructured data.
  - **2024**: Research on LLM Guardrails against attacks and toxicity; Evaluation of LLMs for code generation.
  - **2023**: Zero-shot text classification through different approaches (NLI, semantic similarity, in-context learning), and LLM fine-tuning regimes (PEFT, LoRA, accelerate).

- **Data Science MBA (University of São Paulo)** — Remote
  TA/Tutor — *February 2020 - December 2024*

  - **Responsibilities**: Teaching, Preparing and revising materials, grading, interacting with students.
  - **Classes**: Neural Networks and Deep Learning, Machine Learning, Introduction and Programming for Data Science.

- **The AI Academy** — Remote
  Data Scientist Consultant — *May 2020 - October 2020*

  - **Continuous Development for DS**: Prototyped and evaluated new DS processes and tools.
  - **DevOps**: Integrated H2O Driverless AI into automated CD pipelines for one of Brazil's biggest bank.

- **Imaflora Origens** — Brazil
  Data Scientist — *February 2020 - July 2020*

  - **ETL**: Data wrangling Geo-spatial data from different sources for a interactive visualisation.
  - **Visualisation**: Worked alongside business owners to build a interactive platform for exploring data-driven insights.

- **University of São Paulo, ICMC** — Brazil
  Research Assistant - International scholarship by USAID — *March 2017 - December 2019*

  - **Processing**: Transformed raw audio data (mosquito's wingbeats) into structured data using time series analysis.
  - **Classifying**: Building pipelines and evaluating models to classify different insect species.
  - **Undergratuate thesis**: Applying CNNs to raw audio data for insect classification; Analysing covariate shift in the context of sensor generalization.

- **Birdie** — Brazil
  Data Science Intern — *August 2018 - February 2019*

  - **Data Acquisition and Mining**: Acquired more than 1 million e-commerce products through crawlers, building a reliable back-end infrastructure; Extraction of useful features from the unstructured textual data of the products.
  - **Machine Learning**: Building hierarchical models for product-category classification
  - **Production**: Deployed the models as micro-services processing 10 million+ daily products.

Technical Skills

Languages: **Python**, SQL, C, JavaScript, Java
Topics: NLP, Computer Vision, Deep Learning, MLOps, Big data, Internet Measurement
DS/ML: Pandas, Scikit-Learn, PyTorch, TensorFlow, huggingface, DVC, Seaborn, Plotly
Others: BeautifulSoup, Scrapy, Django, Git, Docker, Airflow, Driverless AI, GCP

Major Projects

- **Winner of the Dotz NLP Competition** May 2020 - July 2020
  - **Competition**: Hierarchical classification of e-commerce products from their description.
  - **Solution**: Building a integrated pipeline using a mixture of portuguese Bert, Character CNNs, traditional sparse BOW model and hierarchical specific models.
- **Finalist of the Data Science Games 2018** May 2018 - September 2018
  - **Qualification Phase**: Analysed the bond market with data provided by BNP Paribas; Predicted customer interest in a bond using diverse data; Final result in the top 30 worldwide from 220 participating universities.
  - **Final**: Used e-commerce data to help predict customer interest in online shopping.
- **Co-founder of DATA - Extracurricular Data Science Group** September 2018 - January 2020
  - **Co-Founder**: Creating and running a extracurricular data science group recognised by the university.
  - **Teaching**: Weekly classes teaching basic Machine Learning concepts and frameworks to a class of 30 students.
- **Computer Science and Data Science Teaching/Volunteering**
  - **NYU's Introduction to CS**: Python programming general grading, Fall 2021.
  - **Invited DS Course Lead**: Heading the DS part of a 3 month course at a startup incubator, Summer 2019.
  - **Python for DS course**: Teaching industry members at the Universidade de São Paulo, Fall 2019.
  - **Mentoring**: High school student, helping with SATs and career questions, 2018.
  - **Programming Tutor**: Taught fundamentals of computer programming to underprivileged communities, 2017.

Awards

- Winner of the NDD Tech Challenge - First place building ML models to classify fiscal documents
- Finalist of the Data Science Games 2018 - top 20, from 400 participating teams across 220 universities and 46 countries.
- Silver medal on Elo's Merchant Category Recommendation on Kaggle. Over 4000 teams of 1-4 people took part.
- 19th place at Data Mining Cup, from 149 Teams from 114 universities in 28 countries - Best team from Brazil.
- Honorable mention at USP's International Symposium of Undergrad Research. 600 mentions given from 3200 applicants.
- Recipient of Academic Featured Student (2016), for students with a GPA above 9 / 10. Awarded to 7 / 120 students.

Invited Events

- **PyData 2023**: Invited participant and scholarship winner.
- **NYU**: Invited speaker at NYU's PhD prospective students and NYU's undergraduate panel in academia (2022, 2023).
- **Khipu 2019**: Latin American Meeting In Artificial Intelligence, invited participant with full merit-based scholarship.
- **Data Day**: Head of the discussion panel "Present and Future of AI"with invited professors and industry members.
- **Mathematics Solutions for Industry Problems Workshop (2020)**: Analysing timber transport data for an NGO.